

Mahindra University Hyderabad
École Centrale School of Engineering
End-Semester Examination (2024 - Batch)

Program: B. Tech.

Branch: All

Year: II

Semester: I

Subject: Probability & Statistics (MA2103)

Date: 04/12/2025

Start Time: 10:00 AM

Time Duration: 3 Hours

Max. Marks: 100

Instructions:

- 1) Each question carries 25 marks.
- 2) All questions are compulsory.
- 3) Please start each answer on a separate page and ensure that you clearly number the responses.
- 4) It is essential to explain each step. Correct outcomes without any description will not be evaluated.

Q 1:

25 marks

- (a) A company is developing an automated spam detection system for their email service. From the analysis of 10,000 historical emails, 30% are spam (denote this class as S) and 70% are legitimate (class L). The system analyzes two features for each email:

- Feature F_1 : Whether the email contains promotional words.
- Feature F_2 : Whether the email contains suspicious links.

The following conditional probabilities based on historical data are provided:

$$P(F_1 = \text{Yes} | S) = 0.80, \quad P(F_1 = \text{Yes} | L) = 0.15,$$

$$P(F_2 = \text{Yes} | S) = 0.75, \quad P(F_2 = \text{Yes} | L) = 0.10.$$

Assume that features F_1 and F_2 are conditionally independent within each class. A new email arrives for classification.

- (i) The email contains promotional words ($F_1 = \text{Yes}$). Calculate the probability that this email is spam given this information. [8 marks]
- (ii) Further analysis reveals that the email does not contain suspicious links ($F_2 = \text{No}$). Calculate the probability that this email is spam given both features ($F_1 = \text{Yes}, F_2 = \text{No}$).

[10 marks]

P.T.O.

- (b) At Mahindra University, a class starts at 8:20 AM. The walking time (in minutes) for students traveling from Hostel Phase II to the Main Building is modeled by the probability density function

$$f(t) = \begin{cases} k(12-t), & 0 \leq t \leq 12, \\ 0, & \text{otherwise,} \end{cases}$$

where k is a constant. A student leaves Hostel Phase II at 8:15 AM.

- (i) Find the value of k . [2 marks]
(ii) Find the probability that the student reaches the classroom by 8:20 AM and therefore receives attendance. [5 marks]

Q 2:

25 marks

- (a) A fair die is rolled 600 times. Using Chebyshev's inequality, find a lower bound for the probability that the number of sixes obtained lies between 80 and 120. [10 marks]
(b) A multiple-choice test has 8 questions, each with 3 answer choices of which only one is correct. A student answers each question by rolling a fair six-sided die: the first option is chosen if the die shows 1 or 2, the second option is selected if it shows 3 or 4, and the third option is chosen if it shows 5 or 6. To earn a distinction, the student must answer at least 75% of the questions correctly. If there is no negative marking, what is the probability that the student earns a distinction? [15 marks]

Q 3:

25 marks

- (a) A large online retailer receives product returns each day, and the number of daily returns follows a Poisson distribution with an average of 25 returns per day. Assume returns on different days are independent. Using the Central Limit Theorem, approximate the probability that the retailer will receive more than 10,100 returns in a 400 day period (Use $\Phi(1) = 0.8413$, where Φ denotes the distribution function of the standard normal distribution). [7 marks]
(b) A bank studies randomly selected customer sessions on its online platform. Let X be the number of transactions attempted in a session and let Y be the number of transaction errors observed in that session. The joint probability mass function of (X, Y) is given by

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 2$	2/9	1/9	1/18
$X = 5$	1/3	1/6	1/9

- (i) Find marginal probability mass functions of X and Y . [4 marks]
(ii) Find $E[X]$, $E[Y]$, $\text{Var}(X)$, and $\text{Var}(Y)$. [8 marks]
(iii) Find $E[XY]$, $\text{Cov}(X, Y)$ and the correlation coefficient $\rho_{X,Y}$. [6 marks]

P.T.O.

Q 4:

25 marks

- (a) Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pdf

$$f_X(x) = \begin{cases} \theta \left(x - \frac{1}{2}\right) + 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

where $\theta \in [-2, 2]$ is an unknown parameter. Consider the estimator

$$\hat{\theta}_n = 12\bar{X} - 6$$

for θ .

- (i) Is $\hat{\theta}_n$ an unbiased estimator of θ ? Justify. [6 marks]
 - (ii) Find the mean squared error (MSE) of $\hat{\theta}_n$. [6 marks]
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from a Poisson distribution with parameter λ .
- (i) Derive the likelihood and log-likelihood functions. [4 marks]
 - (ii) Find the maximum likelihood estimator (MLE) of λ . [3 marks]
- (c) The PCB concentration of a fish caught in Lake Michigan was measured by a technique that is known to result in an error of measurement that is normally distributed with a standard deviation of 0.08 ppm (parts per million). Suppose the results of 10 independent measurements of this fish are:

11.2, 12.4, 10.8, 11.6, 12.5, 10.1, 11.0, 12.2, 12.4, 10.6

Give a 95% confidence interval for the average PCB level of this fish. [Use: $z_{0.025} = 1.96$]

[6 marks]

— End of Question Paper —